Alejandro Cuadrón Lafuente

Research Scientist | Self-improving LLM agents: RL post-training and evaluation

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Experience

Amazon AGI | Applied Scientist II — *Sunnyvale, CA* | Feb 2026 – Present

- **Recursive self-improvement.** Built an RSI system in which models **find their own failures** on agentic tasks and **generate the verified training data that fixes them**: **thousands of verified tasks** across tool calling, software engineering, finance and inventory management, now being integrated into the training of **Amazon's flagship model** (unreleased).
- **Led the first asynchronous agentic RL post-training of a 300B+ MoE model at Amazon AGI**, on hundreds of GPUs, extending Slime and Miles: MoE routing replay with a **10× faster** data path (per-step preparation from ∼4 min to 20 s), **3–9× faster** sparse-attention kernels, **+47%** rollout decode throughput (speculative decoding, fp8 KV cache), and score centering to cancel training–inference drift, with a staleness cap and a durable rollout buffer, for stable training under heavy asynchrony.
- **Led the development of frontier evaluations such as BOBA-Bench**: chaining 20 real coding requests drops ten agents' completion from **95% to 51%**, not from worse code but from mistakes their own tests and reviews miss. One external "that's wrong" lifts it to 59%: agents are **only as reliable as the verification they can reach**.

Amazon AGI | Applied Scientist I — *Sunnyvale, CA* | Jul 2025 – Jan 2026

- **Owned the agentic RL pipeline for Amazon Nova 2 Lite**: SFT data generation for multi-turn tool use, RL environment integration, and quality control of external agentic data.
- **τ^2-Bench-Verified**: found systematic ground-truth errors in the standard tool-use agent benchmark and turned the fixes into training data that brought Nova 2 Lite **on par with Claude Sonnet 4.5**; most of τ^3-bench's **75+ task fixes** come from it.
- **SABER** (first author): showed that errors on state-changing actions cut agent success odds by up to **96%**; built a model-agnostic safeguard with mutation-gated verification, targeted reflection and block-based context cleaning (**+28%** relative).

UC Berkeley, Sky Computing Lab | Research Scholar, advised by Ion Stoica — *Berkeley, CA* | Jan 2024 – Jun 2025

- **Led The Danger of Overthinking**, a 16-author collaboration across UC Berkeley, CMU and ETH Zurich: the first systematic study of reasoning models that keep thinking instead of acting in agentic tasks. Mitigating it improves performance ∼**30% at 43% lower cost**; 175+ citations, covered by IEEE Spectrum.
- **OpenHands** (90k+ GitHub stars): agent delegation for sub-agent workflows, native function calling and reasoning effort for o1, evaluation and runtime infrastructure.
- Co-authored JudgeBench (ICLR 2025) on LLM judges, and HashAttention (ICML 2025), vAttention (ICLR 2026) and vCache (ICLR 2026) on efficient inference.

IBM Research | Student Researcher — *Zurich, Switzerland* | Jul 2023 – Dec 2023

Led Catamaran, a framework for efficient, compliant post-training of ML models, later productized and sold to IBM customers; led the IBM–UC Berkeley collaboration.

Publications

Under review	**BOBA: Why Coding Agents Cannot Be Left Alone**	*first author • ICLR 2027*
ICLR 2026	**vAttention: Verified Sparse Attention**	*co-author*
ICLR 2026	**vCache: Verified Semantic Prompt Caching**	*co-author*
ICLR 2026 W	**SABER: Safeguarding Mutating Steps in LLM Agents**	*first author*
ICML 2025	**HashAttention: Semantic Sparsity for Faster Inference**	*co-author*
ICLR 2025	**JudgeBench: A Benchmark for Evaluating LLM-Based Judges**	*co-author*
arXiv 2025	**The Danger of Overthinking: the Reasoning-Action Dilemma in Agentic Tasks**	*first author • 175+ citations*

Open Source and Competitions

Top 1%	**Open Kernel Challenge (Meta × Paradigm)**: built an RSI framework for GPU kernels; no prior CUDA experience.
75+ fixes merged	**τ^2-Bench-Verified**: corrected the standard tool-use agent benchmark; the source of most τ^3-bench task fixes.

Education

University of California, Berkeley | Graduate studies, Computer Science — *Berkeley* | 2024 – 2025

ETH Zürich | M.Sc. Computer Science, Secure and Reliable Systems — *Zurich* | 2022 – 2024

Universidad Politécnica de Madrid | B.Eng. Computer Engineering — *Madrid* | 2018 – 2022

Awards and Recognition

- **Showed OpenAI overstated o1's agentic coding**: 30% on SWE-bench Verified, not 48.9% (X post); covered by The Decoder (2025).
- Interviewed by **IEEE Spectrum** on overthinking in reasoning models (2025).
- **Top of class**: top 5% at UPM with honors in 9 subjects; top of the class in two ETH Zurich security courses.
- **Excellence scholarships**: Fundación Mutua Madrileña (M.Sc., ETH Zurich) and Government of Madrid (B.Eng., UPM).